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Study on the Catalytic Reduction Performance of Mg-Doped $NiFe_2O_4$ Ferrite for CO_2 by Adopting the Co-Precipitation Method

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Abstract

Spinel ferrites offer a versatile platform for high-temperature CO_2 conversion, yet simultaneously achieving strong adsorption/activation and long-cycle thermal stability remains challenging. Here, we tailor the defect chemistry and microstructure of $NiFe_2O_4$ through low-level A/B-site modification by partially substituting Ni with Mg ($Ni_{0.96}Mg_{0.04}Fe_2O_4$). The catalyst was synthesized by Mg doping and characterized comprehensively by ICP, XRD, SEM and CO_2-TPD, followed by evaluation of CO_2 adsorption and thermal decomposition activity under cyclic operation. Mg incorporation suppresses grain coarsening, refines crystallites, increases accessible surface area and reduces particle size, thereby improving resistance to thermal aging. The enriched oxygen-vacancy population enhances oxygen storage and strengthens CO_2 adsorption, which translates into higher catalytic utilization of active sites. Under repeated CO_2 decomposition cycles, the Mg-modified ferrite shows markedly improved stability and service life, achieving a carbon deposition of 19.62%. The combined evidence indicates that Mg substitution stabilizes the spinel lattice against sintering while promoting vacancy-assisted CO_2 activation, providing a simple and cost-effective compositional lever to balance activity and durability for high-temperature CO_2-to-carbon conversion.

Keywords: ferrite; performance characterization; doping; oxygen storage capacity; grain refinement

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1. Introduction

Decarbonizing high-temperature exhaust streams—exemplified by marine diesel engines—demands catalytic routes that operate reliably under harsh thermal cycling with minimal penalties in footprint and energy consumption. Current practice on ships still favors post-combustion carbon capture (absorption, adsorption, membranes, low-temperature routes), yet the space, energy and safety costs, as well as onboard CO_2 storage, limit adoption in many scenarios [1–9]. Catalytic CO_2 conversion therefore remains an attractive complementary pathway within shipboard and other high-temperature contexts. In parallel, modern CO_2 reduction strategies span photo-, electro-, bio- and thermal catalysis; among these, thermal routes are directly compatible with high-temperature exhaust conditions [9].

From a reaction-engineering perspective, direct CO_2-to-C conversion is strongly endothermic and non-spontaneous, so achieving practical rates requires both activation of the linear, weak-donor/strong-acceptor CO_2 molecule and robust catalysts that tolerate severe temperatures [10]. Mixed-oxide spinels (ferrites) are promising because their defect-tolerant lattices enable concurrent tuning of cation distribution, oxygen-vacancy chemistry and

microstructure—parameters linked to adsorption, redox and sintering resistance under thermal operation.

Elemental substitution in spinel ferrites provides a practical lever to stabilize the framework and control vacancy formation. Prior work shows that selected dopants can reshape redox cycles and adsorption while suppressing grain growth and sintering. For example, Zhao Xuejuan reported that metal-ion doping (Ni^{2+}, Co^{2+}, Cu^{2+}, Zn^{2+}) enhances MFO denitrification via increased surface active sites and dual Mn/Fe redox loops, with the trend Co > Ni ≈ Zn > Cu [11]; Shin et al. analyzed $Ni_xCu_{1-x}Fe_2O_4$ and observed excellent reducibility at higher Ni content and the best CO_2 decomposition at x = 0.5 [12]. More broadly, the activity of oxygen-deficient ferrites scales with the degree of deficiency δ, although excessive deficiency can undermine structural integrity [13,14]. In $NiFe_2O_4$ specifically, Fu Maosheng et al. found that 4% Cr balances structural stability with δ and improves catalytic life and carbon deposition [15]. These precedents motivate low-level Mg substitution as a cost-effective route to refine grains, enrich accessible vacancies and strengthen the spinel lattice against sintering.

Therefore, more effective catalysts should be designed to transform CO_2 into various compounds. Porous crystalline frameworks, such as metal–organic frameworks (MOFs), are promising for use in catalytic CO_2 conversion, owing to their strong CO_2 adsorption capacities, high surface areas, high porosity and chemical compositions, and adjustable active sites [16].

Here, we pursue that strategy by partially replacing Ni with Mg to obtain $Ni_{0.96}Mg_{0.04}Fe_2O_4$ via a scalable co-precipitation route followed by in situ H_2 activation to generate oxygen-deficient phases. This article will focus on the physicochemical and catalytic properties of composite oxide materials formed by partial substitution of Ni ions with alkaline earth metal element Mg. We construct an integrated structure-property-performance map using ICP, XRD, SEM, particle-size analysis, BET surface area, oxygen storage capacity, CO_2-TPD and cyclic CO_2 decomposition tests.

2. Materials and Methods

2.1. The Sources and Purity of All the Chemical Materials

The chemical reagents and materials used for the preparation of Mg-doped $NiFe_2O_4$ are summarized in Table 1. All chemicals were used as received without further purification.

Table 1. Specifications and sources of the chemical reagents used in the experiments.

Chemical	Purity (%)	Source
Ni $(NO_3)_2 \cdot 6H_2O$	99.99	China National Medicines Corporation Ltd., Beijing, China.
Fe $(NO_3)_3 \cdot 6H_2O$	99.99	China National Medicines Corporation Ltd., Beijing, China.
$Mg(NO_3)_2 \cdot 6H_2O$	99.99	China National Medicines Corporation Ltd., Beijing, China.
$C_6H_8O_7 \cdot H_2O$	99.99	China National Medicines Corporation Ltd., Beijing, China.
NaOH solution	4 mol/L	China National Medicines Corporation Ltd., Beijing, China.
Deionized water	99.99	China National Medicines Corporation Ltd., Beijing, China.

2.2. Chemical Reaction Formula of the Preparation

The preparation of the chemical reaction formula is as follows:

$$Ni(NO_3)_2 + 2NaOH = 2NaNO_3 + NiO + H_2O \quad (1)$$

$$2Fe(NO_3)_3 + 6NaOH = 6NaNO_3 + Fe_2O_3 + 3H_2O \quad (2)$$

$$Mg(NO_3)_2 + 2NaOH = 2NaNO_3 + MgO + H_2O \quad (3)$$

2.3. Methods of the Preparation

In this experiment, Ni $(NO_3)_2 \cdot 6H_2O$, Fe $(NO_3)_3 \cdot 6H_2O$ and Mg $(NO_3)_2 \cdot 6H_2O$ were selected as raw materials, and NaOH was used as a precipitant to prepare Mg-doped $NiFe_2O_4$ by mixed ion co-precipitation method.

3. Preparation of Catalyst

The catalyst preparation route critically impacts texture, defect chemistry and activity. Prior studies on ferrite and ferrite-based materials have explored solid-state synthesis, co-precipitation (including microwave-assisted variants), hydrothermal routes and precursor decomposition for photocatalysis or oxidation processes, providing practical guidance on parameter windows and post-treatments [17–23]. Guided by these reports and aiming at scalable processing, we adopted a co-precipitation method to prepare $NiFe_2O_4$ and Mg-modified $Ni_{0.96}Mg_{0.04}Fe_2O_4$.

There is an electron transfer mechanism between Mg and Fe/Ni, mainly achieved through the electric couple effect and interface electron tunneling, which is manifested as accelerating electron transfer and reducing the reaction energy barrier.

In the hydrolysis reaction between $NiFe_2O_4$ catalyst and Mg, $NiFe_2$ acts as the active site and undergoes electronic coupling with Mg through the electric couple effect. The electrons on the surface of $NiFe_2$ rapidly transfer to the hydrogen ions (H^+) generated during the hydrolysis reaction of Mg, forming a low-energy barrier electron migration path and significantly improving the hydrolysis rate.

3.1. Preparation of $NiFe_2O_4$ and $Ni_{0.96}Mg_{0.04}Fe_2O_4$

Ni $(NO_3)_2 \cdot 6H_2O$, Fe $(NO_3)_3 \cdot 6H_2O$, and, where applicable, Mg $(NO_3)_2 \cdot 6H_2O$, were used as precursors; NaOH served as the precipitant. For $NiFe_2O_4$, Ni^{2+} and Fe^{3+} were dissolved in deionized water at a molar ratio C(Ni^{2+})/C (Fe^{3+}) = 1/2, yielding a clear solution. Under stirring, 2M (mol/L) NaOH was added dropwise at 60–80 °C to adjust pH = 10–12, promoting co-precipitation. After 2–4 h aging, the slurry was centrifuged; the solid was repeatedly washed with deionized water (to near-neutral filtrate) and rinsed with ethanol to remove residual Na^+ and soluble impurities. The precipitate was dried at 60–80 °C, then calcined in air at 800–1000 °C for 4–6 h to obtain $NiFe_2O_4$ with spinel structure.

For $Ni_{0.96}Mg_{0.04}Fe_2O_4$, the same protocol was employed but the total (Ni^{2+} + Mg^{2+}) to Fe^{3+} concentration ratio was maintained at 1/2, and $Mg(NO_3)_2 \cdot 6H_2O$ was introduced to achieve 4% Mg substitution on the Ni sites. The resulting Mg-modified sample was processed identically (washing, drying and calcination) to ensure comparability.

3.2. Preparation of $NiFe_2O_{4-\delta}$ and $Ni_{0.96}Mg_{0.04}Fe_2O_{4-\delta}$

Following Fu Maosheng et al. [15], oxygen-deficient $NiFe_2O_{4-\delta}$ was obtained by H_2 reduction at 310 °C with a hydrogen flow of 40 mL·min^{-1} for 3 h, which produced $\delta > 0$ with high structural stability and activity. Excessive temperature (≥400 °C) or H_2 flow (e.g., 160 mL·min^{-1}) can induce spinel collapse and formation of Fe_xNi_{1-x} alloy and α-Fe,

thereby degrading performance [15]. For $Ni_{0.96}Mg_{0.04}Fe_2O_4$, improved bond stability supports a slightly higher optimal reduction temperature of 320 °C (other conditions unchanged). In practice, a fixed-bed in a temperature-controlled reactor was used: after the H_2 reduction step, the reactor was evacuated for 20 min, purged with Ar for 20 min, and evacuated again for 10 min before introducing CO_2 and balance gas for catalytic testing (see Equations (4) and (5) for the vacancy-formation stoichiometry). The same sequence was applied to both compositions to ensure consistent pretreatment. The reaction formula for oxygen-vacancy preparation is as follows:

$$NiFe_2O_4 + \delta H_2 = NiFe_2O_{4-\delta} + \delta H_2O \tag{4}$$

$$Ni_{0.96}Mg_{0.04}Fe_2O_4 + \delta H_2 = Ni_{0.96}Mg_{0.04}Fe_2O_{4-\delta} + \delta H_2O \tag{5}$$

4. Characterization of Catalysts

Inductively coupled plasma (ICP) analysis was performed to validate the bulk stoichiometry against the nominal compositions and to ensure that co-precipitation/calcination did not introduce compositional deviation during Mg substitution. ICP composition analysis was conducted on the sample powder, and the test results are shown in Table 2.

Table 2. ICP test results of sample.

Feeding Ratio	ICP Test Results
Ni:Fe = 1:2	Ni:Fe = 0.97:1.99
Ni:Fe:Mg = 0.96:2:0.04	Ni:Fe:Mg = 0.94:2.1:0.042

ICP test results show that the actual composition is Ni:Fe:Mg = 0.94:2.1:0.042, Mg doping ratio (0.042) is closest to the theoretical value (0.04), while Ni and Fe have relative deviations of −2.1% and +5.0%, respectively. Based on the analysis of potential factors such as reaction condition control, precipitation separation efficiency and testing errors, it is indicated that the deviation is mainly caused by small fluctuations in pH, temperature and other parameters during co-precipitation, as well as operational errors in ICP sample digestion or dilution. Despite local deviations, the overall composition still meets the expected structure of $Ni_{0.96}Mg_{0.04}Fe_2O_4$, demonstrating the feasibility of this method in doping control.

X-ray diffraction (XRD) was conducted to confirm the target crystalline phase and to assess lattice perturbations or secondary phases associated with Mg incorporation, thereby verifying the structural integrity of the perovskite. From Figure 1, it can be seen that the main peak positions of the two samples are consistent with the standard spectrum of $NiFe_2O_4$ spinel, and no obvious crystalline impurity phases are detected. Only several very weak reflections are observed at approximately 19°, 33° and 67°, which are most likely attributed to trace surface/by-product species and/or instrumental background. These results indicate that Mg^{2+} has entered the lattice structure of $NiFe_2O_4$. Due to the smaller radius of Mg^{2+} (0.066 nm) compared to Ni^{2+} (0.069 nm), there will be changes in the lattice constant. This lattice distortion causes the diffraction peaks to shift towards higher angles. From the figure, it can be seen that the peak intensity of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ is significantly higher than that of $NiFe_2O_4$, and the characteristic peak becomes very sharp. This indicates that the grain crystallinity has been improved, which is generally beneficial for the thermal and structural stability of the catalyst.

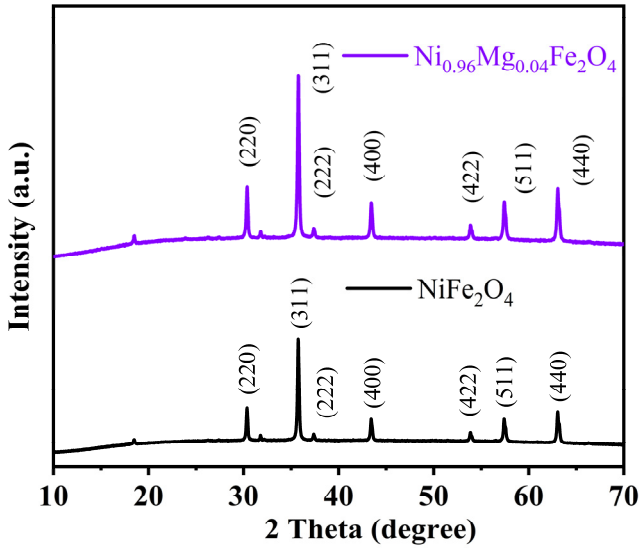

Figure 1. XRD spectra of $NiFe_2O_4$ before and after doping.

According to the XRD average grain parameter test results in Table 3, the average grain size of the sample decreases due to doping. The literature value of the unit cell parameters of $NiFe_2O_4$ is 0.834 nm. Due to the different radius sizes, when Ni^{2+} in $NiFe_2O_4$ is replaced by Mg^{2+}, this will lead to a decrease in the unit cell parameters of $NiFe_2O_4$. It can be inferred from the reduced doping parameters that the Mg^{2+} doped in the sample has successfully entered the lattice of $NiFe_2O_4$.

Table 3. Average grain size and unit cell parameter of samples.

Samples	Average Grain Size/nm	Unit Cell Parameter/nm
$NiFe_2O_4$	26.6	0.83377
$Ni_{0.96}Mg_{0.04}Fe_2O_4$	22.4	0.83296

Scanning electron microscopy (SEM) with energy-dispersive X-ray mapping (EDS) was employed to resolve particle morphology, grain connectivity and the spatial distribution of Mg relative to host cations, thereby evaluating whether Mg substitution suppresses overgrowth or agglomeration and whether locally enriched domains appear. Complementary particle-size analysis and BET surface-area measurements quantify the resulting textural changes, establishing the morphological basis for subsequent CO_2 adsorption performance and high-temperature durability.

The SEM/EDS analysis of pristine From $NiFe_2O_4$ is illustrated in Figure 2. In comparison, it can be seen that the Mg element is evenly distributed in $NiFe_2O_4$, and the element amount is relatively small, which conforms to the chemical formula $Ni_{0.96}Mg_{0.04}Fe_2O_4$ (Figure 3). The distribution of particles doped with Mg is more uniform, without agglomeration, and there are more pores with overlapping particles, which gives it a larger surface area. Mg doping can effectively inhibit the aggregation of $NiFe_2O_4$, providing a larger surface area, which means that more CO_2 can be adsorbed on the surface and the catalytic performance can be improved.

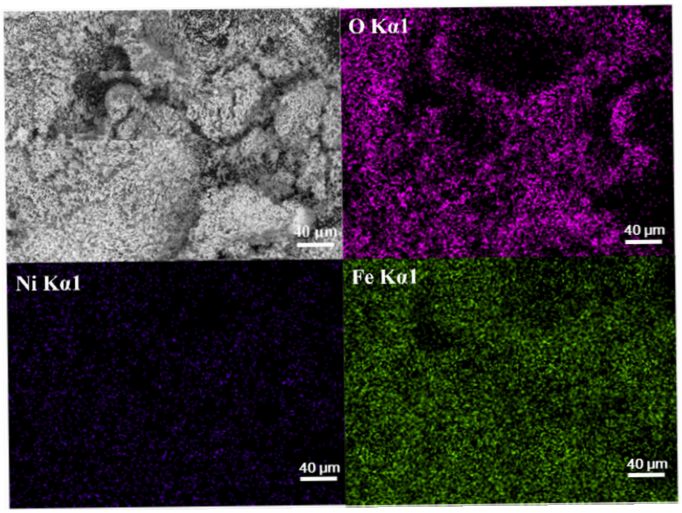

Figure 2. SEM and EDS images of $NiFe_2O_4$. The EDS maps illustrate the distribution of oxygen (O, purple), nickel (Ni, blue), and iron (Fe, green).

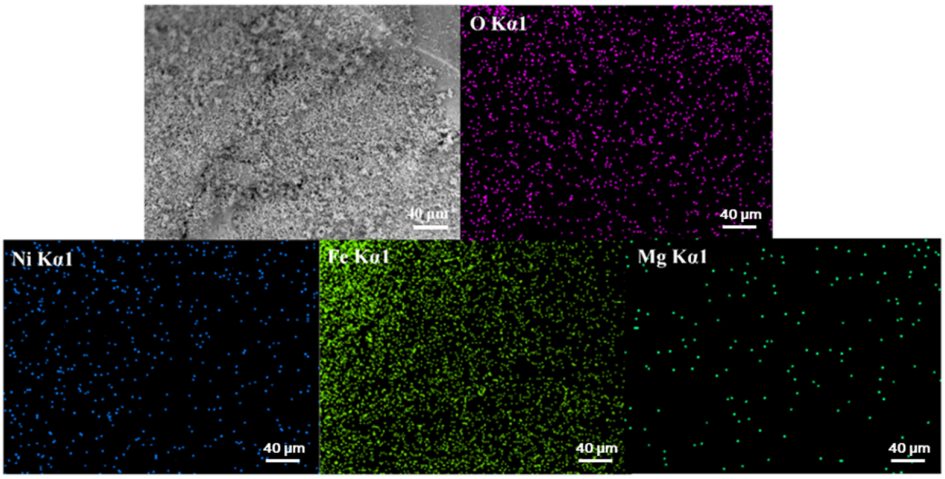

Figure 3. SEM and EDS images of $Ni_{0.96}Mg_{0.04}Fe_2O_4$. The EDS maps illustrate the distribution of oxygen (O, purple), nickel (Ni, blue), iron (Fe, green), and magnesium (Mg, light green).

By comparing Figure 4a,b, it can be seen that sintering and agglomeration phenomena occur after thermal aging, and the particle size distribution is not very uniform. Large and small particles coexist, indicating that the grains do not grow uniformly during the growth process. Comparing Figure 5a,b, it can be seen that there is only a small amount of sintering phenomenon before and after thermal aging, and the grains grow uniformly without agglomeration. The surface still retains fine grains and has a large surface area. The presence of a small amount of sintering phenomenon may be due to the uneven distribution of Mg, which leads to sintering in this area. This also indicates that Mg doping can indeed refine the grain size and prevent grain growth and sintering.

The specific surface area can reflect the degree of contact between the catalyst powder and the outside world. Multiphase reaction is a surface reaction, and the surface active component is the active center of the reaction. Chemical adsorption and redox reactions occur on the surface of powder particles. A larger specific surface area can optimize the contact with reactants, enhancing the adsorption and catalytic performance of the catalyst. The particle size and specific surface area of $NiFe_2O_4$ with different particle sizes before and after doping were measured using a laser particle size analyzer and a specific surface area analyzer, as shown in Table 4.

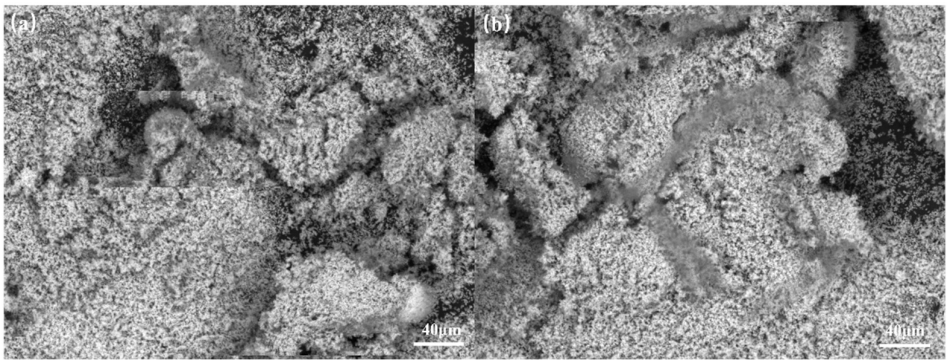

Figure 4. Comparison of $NiFe_2O_4$ before and after thermal aging. (**a**) $NiFe_2O_4$ before thermal aging; (**b**) $NiFe_2O_4$ after thermal aging.

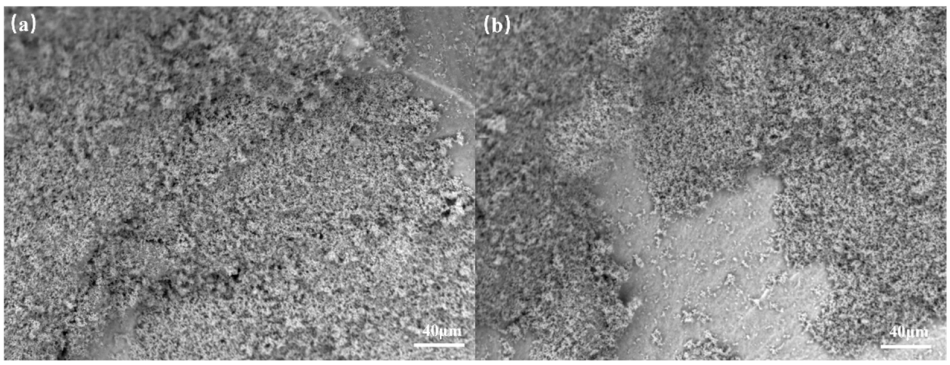

Figure 5. Comparison of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ before and after thermal aging. (**a**) $Ni_{0.96}Mg_{0.04}Fe_2O_4$ before thermal aging; (**b**) $Ni_{0.96}Mg_{0.04}Fe_2O_4$ after thermal aging.

Table 4. Particle size and specific surface area before and after Mg doping.

Catalyst	D10 (μm)	D50 (μm)	D90 (μm)	Average Particle Size Reduction Rate (%)	Specific Surface Area (m^2/g)	Specific Surface Area Improvement Ratio (%)
$NiFe_2O_4$	1.26	4.13	8.74	0	52.78	0
$Ni_{0.96}Mg_{0.04}Fe_2O_4$	0.92	2.97	6.45	28.1	69.57	31.8

By comparing the data in Table 4, it was found that under the same preparation conditions, the particle size of each particle size grade was significantly reduced after Mg partially replaced Ni, especially the average particle size D50 of the reaction. The D50 of the catalyst doped with Mg was 2.97 μm, which was 28.1% lower than that of the undoped condition, and the specific surface area was also improved. After adding 4% Mg to the catalyst, the catalyst grains became smaller during the formation process and did not continue to grow. This may be because Mg alleviates agglomeration, thereby hindering grain growth. After doping with Mg, the specific surface area of the catalyst powder reached 69.57 m^2/g, an increase of 31.8% compared to the undoped catalyst.

Generally speaking, there is an inverse correlation between the surface area and particle size of particulate matter. However, in this experiment, the decrease in particle size was significantly less than the increase in specific surface area. The reason is that specific surface area refers to the total area per unit mass of material. Because when the particle size decreases, the number of particles per unit mass or volume increases, resulting in a significant increase in total surface area. This law has important application value in fields such as nanomaterials and catalysts.

It can be inferred that the growth rate of the surface area inside the pores of the particles exceeds the growth rate of the external surface area, leading to an increase in porosity. The increase in porosity effectively prevents the sintering phenomenon on the surface of catalyst powder, which helps to construct and improve the pore structure.

In summary, Mg plays a positive role in refining grain size, inhibiting agglomeration, and high-temperature sintering, resulting in larger specific surface area and higher porosity of the prepared powder. This is consistent with the analysis results in SEM.

5. Test Experiment of Oxygen Storage Performance and Anti-Aging Performance

We measure oxygen storage capacity at an application-relevant temperature to gauge the availability of redox-active oxygen vacancies in fresh catalysts. This baseline enables a direct comparison between undoped and Mg-modified samples and provides a quantitative bridge to CO_2 adsorption/activation in later sections. Testing steps: the sample was placed into the reactor, H_2 was introduced to it. The temperature of the reactor was raised to 310 °C~320 °C, the powder sample was reduced for 3 h, then N_2 was used to purge, the sample was cooled to 200 °C, pulse O_2 was introduced to the reactor at constant temperature, and the oxygen adsorption capacity was measured. Based on the oxygen adsorption capacity, the oxygen storage capacity of the sample was calculated.

The test results of oxygen storage capacity of $NiFe_2O_4$ before and after Mg doping are as follows: before being doped, the oxygen storage capacity of $NiFe_2O_4$ is 420 μmol/g, and after being doped, that of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ reaches 452 μmol/g, which is 32 μmol/g higher than the former, with an increase of 7.6%. Table 5 reflects the changes before and after being doped.

Table 5. Changes in oxygen storage before and after Mg doping.

Catalyst	Specific Surface Area (m^2/g)	Specific Surface Area Improvement Ratio (%)	Oxygen Storage Capacity (200 °C) (μmol/g)	Growth Rate of Oxygen Storage Capacity (%)
$NiFe_2O_4$	52.78	0	420	0
$Ni_{0.96}Mg_{0.04}Fe_2O_4$	69.57	31.8	452	7.6

The oxygen storage performance of ferrite materials comes from the reversible conversion between Fe^{2+} and Fe^{3+} [24], where gaseous oxygen is adsorbed into lattice oxygen. After Mg doping, the specific surface area of the catalyst increases, the contact area between the powder and the gas and reactants increases, and the adsorption capacity is improved. At the same time, doping ions enter the lattice, causing lattice defects, increasing oxygen vacancies, improving lattice oxygen mobility, and enhancing oxygen storage capacity.

Thermal robustness is probed by subjecting catalysts to severe aging and then comparing property retention against fresh baselines. The aging of catalyst powder is mainly manifested in changes in the surface structure and bulk structure of the powder. Under high-temperature conditions, the powders are prone to sintering and agglomeration, resulting in a decrease in specific surface area, a reduction in surface active sites, a deterioration of the contact interface with reactants and a decrease in adsorption and catalytic performance. Meanwhile, high temperature can also cause changes in the phase structure of the catalyst, severely affecting its activity or even completely losing it. The ferrite composite oxide has better anti-aging stability, and can still maintain good oxygen storage performance at about 1200 °C [25].

The catalyst aging test method is to put a small amount of sample powder under a high temperature of 1200 °C for aging for 4 h, then it was cooled to room temperature, and its specific surface area and oxygen storage capacity were measured to evaluate the anti-aging performance. The specific parameters are shown in Table 6.

Table 6. Test values of specific surface area and oxygen storage capacity after thermal aging.

Catalyst	Aging Temperature (°C)	Specific Surface Area (m^2/g)	Specific Surface Area Reduction Ratio (%)	Oxygen Storage Capacity(200 °C) (μmol/g)	Reduction Ratio of Oxygen Storage Capacity (%)
$NiFe_2O_4$	1200 (4 h)	18.83	64.3	248	41.0
$Ni_{0.96}Mg_{0.04}Fe_2O_4$	1200 (4 h)	26.43	62.0	343	24.1

The experimental results indicate that the surface of the sample powder will undergo varying degrees of sintering and agglomeration under high-temperature conditions, resulting in a decrease in the specific surface area and oxygen storage capacity of the powder. After thermal aging, the specific surface area of the ferrite powder decreased by 33.95 m^2/g compared to before aging, with a reduction rate of 64.3%. The oxygen storage capacity decreased by 172 μmol/g, with a reduction rate of 41.0%; The specific surface area of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ decreased by 43.14 m^2/g compared to before aging, with a reduction rate of 62.0%. The oxygen storage capacity decreased by 109 μmol/g, with a reduction rate of 24.1%.

It is generally believed that the smaller the decrease in specific surface area and oxygen storage capacity of the catalyst after aging, the better its resistance to high-temperature aging. Therefore, the doping of Mg to some extent inhibits the surface sintering of catalyst powder at high temperatures, reduces the influence of high temperatures on specific surface area and oxygen storage capacity, and improves the high-temperature aging resistance of the catalyst.

6. Testing Experiments of Gas Adsorption Performance and Catalytic Activity

6.1. Testing Experiments of Gas Adsorption Performance

In multiphase catalytic reactions, the adsorption capacity of the catalyst powder surface for the reaction gas has an impact on the entire catalytic reaction process and can determine the start of the catalytic reaction. The pre-treated catalyst powder was subjected to a CO_2 temperature-programmed desorption (CO_2-TPD) experiment in a temperature-controlled reactor to evaluate its CO_2 adsorption behavior. An amount of 50 g of ground catalyst was evenly spread on a stainless steel plate inside the temperature controller reactor. First, 6 L/min argon (Ar) blew for 1 h at 400 °C and removed the adsorbed gases and moisture on the surface of the powder. Subsequently, the temperature was lowered to 200 °C in an Ar atmosphere, and Ar was replaced with a standard gas containing 1000 ppm CO_2 + He, and the temperature was programmed to 500 °C. The CO_2 concentration value was measured every minute, and the CO_2 adsorption performance was measured by the CO_2 adsorption rate at different temperatures. Figure 6 showed the variation curve of CO_2 concentration in the CO_2 adsorption experiment before and after Mg doping.

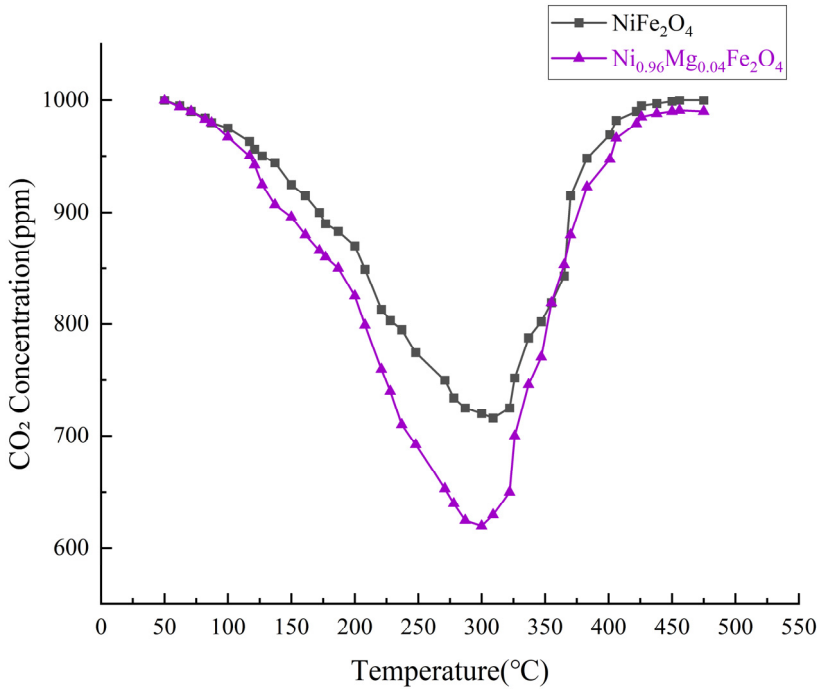

Figure 6. CO_2-TPD profiles of $NiFe_2O_4$ and $Ni_{0.96}Mg_{0.04}Fe_2O_4$.

The decrease in CO_2 concentration represents the adsorption capacity of CO_2 on the catalyst surface. The adsorption rate is calculated by measuring the CO_2 concentration at the outlet to evaluate the adsorption performance of the catalyst.

$$[CO_2]_a = 1 - \frac{[CO_2]_m}{1000} \tag{6}$$

In the Formula (6), $[CO_2]_m$ represents the measured value of CO_2 concentration at the outlet of the temperature controlled reactor. $[CO_2]_a$ represents the absorption rate of the catalyst.

From the changes in the CO_2 concentration curve in Figure 6, it can be observed that there are two distinct concave regions, indicating that the adsorption process can be divided into two stages: the low-temperature stage (100–200 °C) and the high-temperature stage (200–300 °C).

In the low-temperature stage, catalyst adsorption usually manifests as physical adsorption of CO_2, with relatively low adsorption rate and amount, mainly caused by surface diffusion and viscosity. The adsorption curve of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ shows a significant downward depression, and the degree of depression is greater than that of $NiFe_2O_4$. This is mainly because $Ni_{0.96}Mg_{0.04}Fe_2O_4$ has a larger specific surface area and better surface structure, which increases the contact area between the powder and CO_2, thereby enhancing the physical adsorption capacity of $Ni_{0.96}Mg_{0.04}Fe_2O_4$.

During the high-temperature stage, the catalyst is activated and begins to chemically adsorb CO_2. The adsorption curve of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ is lower than that of $NiFe_2O_4$, indicating a faster CO_2 adsorption rate and an increase in CO_2 adsorption at each temperature point. At the same time, the adsorption peak significantly shifted downwards, and the CO_2 adsorption rate reached 38.1% at the highest peak. The adsorption strength and amount of CO_2 by Mg-doped $NiFe_2O_4$ were significantly improved.

After partially replacing Ni with Mg, due to the difference in atomic radius between Mg and Ni and the potential balance after Mg^{2+} infiltration, there will be more structural defects and oxygen vacancies. Therefore, it indicates that Mg doping can indeed provide

more CO_2 adsorption sites for catalysts. The activity, selectivity and stability of the doped sample are enhanced, as shown in Table 7.

Table 7. Changes in surface characteristics of the doped sample.

Catalyst	Activity	Selectivity	Stability
$NiFe_2O_4$	General	General	General
$Ni_{0.96}Mg_{0.04}Fe_2O_4$	Enhance	Enhance	Enhance

Figure 7 shows the variation curve of CO_2 concentration with balanced pressure in the CO_2 adsorption experiment after Mg doping at four different temperatures (50/100/150/200 °C). As shown in Figure 7, as the balanced pressure increases, the CO_2 concentration decreases in the four different temperatures. This indicates that the activation performance of the catalyst improves with increasing temperature and balanced pressure.

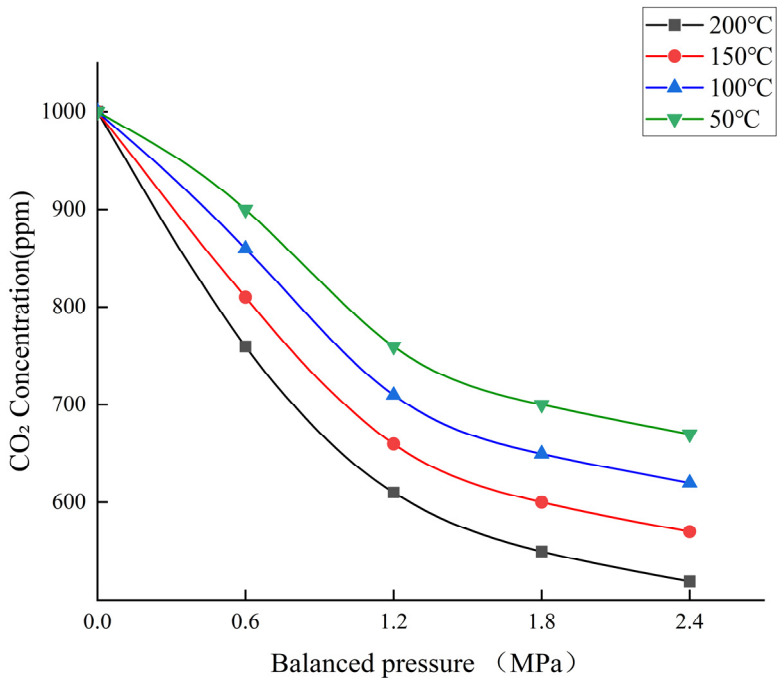

Figure 7. CO_2 concentration changes of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ at four different temperatures.

The commonly used empirical formula for calculating the activation energy of a catalyst is the Arrhenius equation, which describes the relationship between the reaction rate constant and the reaction temperature. The specific formula is as follows:

$$K = A \times exp(-Ea/RT) \tag{7}$$

In the formula, K is the reaction rate constant, A is the pre-exponential factor, Ea is the activation energy, R is the ideal gas constant and T is the reaction temperature.

6.2. Testing Experiments of Catalytic Activity

A temperature-controlled reactor was used as a catalytic reaction platform to evaluate catalyst activity. A total of 20 g of catalyst was loaded into the reactor, and under a hydrogen atmosphere, oxygen-deficient catalysts ($NiFe_2O_{4-\delta}$, $Ni_{0.96}Mg_{0.04}Fe_2O_{4-\delta}$) were prepared using a reduction temperature of 310 °C, a hydrogen flow rate of 40 mL·min^{-1} and a reduction time of 3 h. After the reduction was completed, the reaction chamber was evacuated for 20 min using a vacuum pump, then purged with argon Ar for 20 min, and

finally evacuated for 10 min using a vacuum pump. At this time, CO_2 was introduced to 0.1 MPa for CO_2 activity evaluation experiments, and P-t data was immediately recorded. The total reaction time was 2–4 h. When the number of cyclic reactions reaches a certain value, CO_2 cannot be completely reduced. Before starting the next cycle reaction, it is necessary to evacuate the remaining gas in the reaction system and then introduce H_2 for the next round of reduction reaction. After the cyclic reaction is completed, the sample is cooled to room temperature, then removed, and its carbon deposition is detected using an elemental analyzer. The specific reaction Equations (8)–(11) involved in this process are as follows:

$$NiFe_2O_4 + \delta H_2 = NiFe_2O_{4-\delta} + \delta H_2O \tag{8}$$

$$Ni_{0.96}Mg_{0.04}Fe_2O_4 + \delta H_2 = Ni_{0.96}Mg_{0.04}Fe_2O_{4-\delta} + \delta H_2O \tag{9}$$

$$NiFe_2O_{4-\delta} + \frac{\delta}{2}CO_2 = NiFe_2O_4 + \frac{\delta}{2}C \tag{10}$$

$$Ni_{0.96}Mg_{0.04}Fe_2O_{4-\delta} + \frac{\delta}{2}CO_2 = Ni_{0.96}Mg_{0.04}Fe_2O_4 + \frac{\delta}{2}C \tag{11}$$

Figure 8 shows the variation curve of CO_2 pressure with the number of cycles in $NiFe_2O_4$ cyclic reaction before and after Mg doping. In the first and second cycles of the reaction, the internal pressure of CO_2 drops to zero, indicating that CO_2 is completely converted to C. As the number of cycles increases, the pressure change of CO_2 shows a step-like decrease, which means that the ability of oxygen vacancies for $NiFe_2O_{4-\delta}$ to decompose CO_2 also shows a step-like decrease.

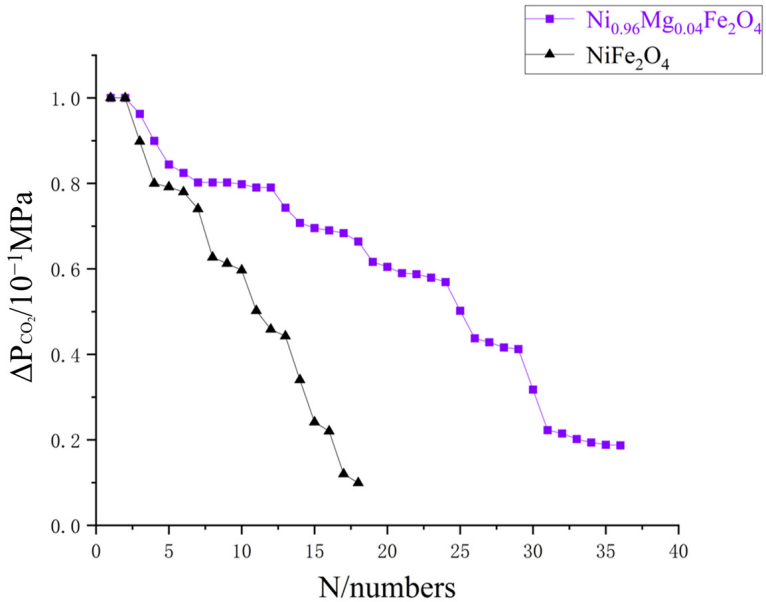

Figure 8. Variation curve of CO_2 pressure with the number of cycles in $NiFe_2O_4$ cyclic reaction before and after Mg doping.

For $Ni_{0.96}Mg_{0.04}Fe_2O_4$, as the number of cycles increases, the CO_2 pressure change also shows a step-like decrease, but the rate of decrease in CO_2 conversion is much slower than that of $NiFe_2O_4$, which means its activity decreases slower and can withstand more cycles of reaction.

In the first three cycles of the reaction, high reaction activity is observed, with almost all of CO_2 converted to C. After five cycles, the conversion rate of CO_2 decreases to around 82%. As the number of cyclic reactions continues to increase, the catalytic activity maintains a relatively long stability. When the number of cyclic reactions reaches 12, the CO_2 conversion rate remains at around 80%, while the CO_2 conversion rate of $NiFe_2O_4$ is only about 46%.

After 29 cycles of reaction, the CO_2 conversion rate is around 41%, and the catalytic activity quickly decreased thereafter. After 36 cycles of reaction, the CO_2 conversion rate is only about 20%, indicating low decomposition activity. From this, it can be seen that Mg doping in $NiFe_2O_4$ improves its stability, significantly slowing down the rate of decomposition activity reduction after cyclic reactions, greatly increasing its cycle reaction frequency and service life.

After the cyclic reaction is completed, the carbon deposition is detected by an elemental analyzer. From the data in Table 8, it can be seen that after 4% Mg doping, the cyclic reaction number and carbon deposition of $NiFe_2O_4$ have been greatly improved and enhanced. The maximum number of cyclic reactions for $Ni_{0.96}Mg_{0.04}Fe_2O_4$ is 36 times, with a maximum carbon deposition of 3.92 g, accounting for 19.62%. Meanwhile, $NiFe_2O_4$ has a maximum number of cyclic reactions of only 18 times, with a maximum carbon deposition of 2.10 g, accounting for only 10.52%.

Table 8. Cycle reaction times and carbon deposition of $NiFe_2O_4$ before and after Mg doping.

Sample (20 g)	Maximum Number of Cyclic Reactions (Time)	Maximum Carbon Deposition (g)/Accounting (%)
$NiFe_2O_4$	18	2.10/10.52
$Ni_{0.96}Mg_{0.04}Fe_2O_4$	36	3.92/19.62

Figures 9 and 10 show the XRD patterns of $NiFe_2O_4$ and $Ni_{0.96}Mg_{0.04}Fe_2O_4$ at different reaction cycles. After increasing the number of cyclic reactions, the characteristic peaks of Fe_xNi_{1-x} and Fe_3C gradually appeared, and the spinel type characteristic peaks of $NiFe_2O_4$ and $Ni_{0.96}Mg_{0.04}Fe_2O_4$ are gradually weakened. This indicates a decrease in the proportion of spinel type phases in the system until they are completely dissolved. Under 10 cycles of reaction, the peak intensities of Fe_xNi_{1-x} and Fe_3C in $NiFe_2O_4$ are significantly higher than those in $Ni_{0.96}Mg_{0.04}Fe_2O_4$, while the peak intensity of spinel type structure is much lower than that in $Ni_{0.96}Mg_{0.04}Fe_2O_4$. This further proves that Mg doping improves the structural stability of catalysts, and its oxygen-deficient structure is less prone to collapse, inhibiting the disintegration rate of spinel type structure and prolonging the cycle reaction times, thereby increasing carbon deposition.

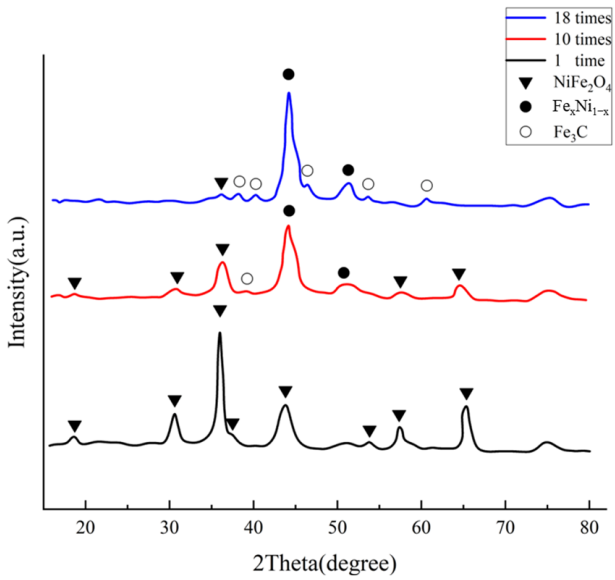

Figure 9. XRD patterns of $NiFe_2O_4$ at different cycle reaction times.

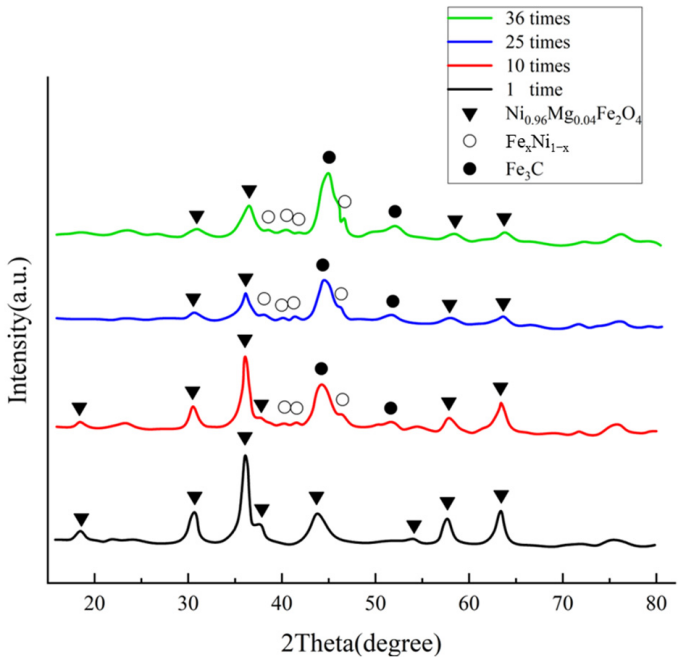

Figure 10. XRD patterns of $Ni_{0.96}Mg_{0.04}Fe_2O_4$ at different cycle reaction times.

7. Conclusions

This paper sought to mitigate the persistent trade-off between activity and durability in thermal CO_2 decomposition through minimal compositional modification of $NiFe_2O_4$. Substituting a small fraction of Ni with Mg and introducing oxygen deficiency via in situ H_2 activation. The rationale was that this low-level compositional change, implemented by a scalable co-precipitation route, could refine grains, enrich accessible vacancies and stabilize the spinel framework against high-temperature sintering, thereby strengthening CO_2 adsorption/activation and sustaining catalytic output. Experiments confirm these expectations. With Mg incorporation ($Ni_{0.96}Mg_{0.04}Fe_2O_{4-\delta}$), the average crystallite size decreased from 26.6 to 22.4 nm and the unit-cell parameter from 0.83377 to 0.83296 nm, the BET surface area increased from 52.78 to 69.57 m^2/g, while the median particle size D_{50} decreased from 4.13 to 2.97 μm. The oxygen storage capacity at 200 °C rose from 420 to 452 μmol/g. After aging at 1200 °C for 4 h, the Mg-doped catalyst showed smaller losses than the undoped material (BET loss 62.0% versus 64.3%; OSC loss 24.1% versus 41.0%), consistent with SEM evidence of suppressed particle coalescence. Temperature-programmed measurements revealed stronger CO_2 uptake with a peak adsorption rate of 38.1%. In cyclic CO_2 decomposition, the operational life increased from 18 to 36 cycles and the maximum carbon deposition from 10.52% to 19.62%. Ex situ XRD during cycling showed delayed formation of Fe_xNi_{1-x} and Fe_3C and better retention of spinel reflections for the Mg-modified sample, indicating improved lattice integrity.

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Data Availability Statement: All data, models, and code generated or used during the study appear in the submitted article.

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